

Nithish Athilingam

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Work Experience:

Northrop Grumman - Associate Software Engineer

February 2025 – Present

- Spearheaded the integration of PowerBI embedding into the MSBI portal, expanding platform capability to support multiple BI providers, enabling teams to seamlessly access additional dashboards on a unified page, by designing a hybrid C#/.NET and Web-Component layout that renders dashboards based on automatic URL-domain detection.
- Developed a secure, high-reliability email routing and real-time user notification system, eliminating third-party mailer dependencies while achieving 100% SMTP RFC-5322 compliance and full unit-test coverage, by implementing a custom low-level SMTP socket.
- Hardened crypto boundaries for 7 services and enabled hardware-rooted data lifecycle controls by integrating TPM 2.0 (tpm2-tss) into an internal C++ framework for sealing/attestation after OS boot.
- Derived and baselined the SRS from a software design review, decomposing stakeholder needs into >30 software requirements and managing them in IBM DOORS with bidirectional traceability; enabled clear verification planning.
- Reduced post-merge regressions by building/maintaining Google Test suites and wiring them into Jenkins Pipeline.

Northrop Grumman - Software Engineering Intern

May 2024 – August 2024

- Delivered a B-52 simulation sustaining ≥60 FPS on target hardware by developing tools/UI in C# (WPF .NET) with XAML/XML data pipelines and scenario development in UE4, enabling faster scenario iteration.

Battelle - Software Engineering Intern

August 2023 – May 2024

- Improved UI responsiveness to ≤10 ms refresh rate by updating an embedded/mobile web app front end in React.js and building Rust/Python WebSocket back-end tests.
- Unblocked development 6 months early by emulating a production-grade gimbal from public ICDs—Go on Raspberry Pi + React front end—accessible on the private Battelle network for rapid testing.
- Aligned cross-team interfaces and cut integration rework by co-authoring Interface Control (ICD) and Requirements documentation for multiple program services.

AdVon Commerce - Data Science Intern

May 2023 – August 2023

- Increased content throughput by 92% and generated article suggestions by building an NLP recommender in Python, containerized with Docker, and deployed on a GCP VM for real-time and batch jobs.
- Surfaced merchandising insights across 100s of products by applying tensor matching and clustering to identify seasonal demand and search-to-sale paths, informing campaigns that drove a lift in revenue.
- Reduced manual product pairing effort and improved CTR by 5% by training TensorFlow/PyTorch image-similarity models to match related products for advertising.
- Built an ingestion pipeline for legacy publishers lacking APIs/DBs, using Selenium (undetected-chromedriver), BeautifulSoup, regex, and pandas to parse messy HTML into a normalized product schema; onboarded >10,000 sources and expanded catalog coverage while cutting manual data entry completely.

Projects:

EcoTech Framework (Sponsored by Honda)

August 2024 – December 2024

- **Publication:** medium.com/99p-labs/the-ecotech-framework-fa531ff293ef
- Built a real-time sustainability dashboard (Python + React, WebSockets, Docker) to measure AI training energy/CO₂ locally and in cloud, using Eco2AI for hardware telemetry and a client-server tracker pipeline.
- Designed a grading model that converts efficiency into an A–F sustainability score by computing CO₂ per FLOPs, then mapping to percentiles (right-skew aware) to fairly compare runs across hardware/regions.

Skills & Certifications:

- **Programming Languages:** Python, C++, C, C#, Java, JavaScript, TypeScript, SQL, MATLAB, GO, x86-64 Assembly
- **Collaboration and DevOps:** Jira, Bitbucket, Confluence, GitHub, GitLab (Agile workflow, code reviews)
- **Frameworks and Libraries:** Next.js, React.js, Node.js, TensorFlow, PyTorch, pandas, Selenium, MonoGame, WPF (.NET), XML/XAML, GTest, TPM2-tss, OpenSSL, Live555, BeautifulSoup, regex, Jenkins
- **OS, Platforms and Tools:** Linux, Windows, Docker, Podman, GCP, Azure, Git, Bash, PowerShell, LTspice, Postman
- **Certifications:** Microsoft Azure Fundamentals, Bloomberg Market Concepts, Associate EDI for X12

Education:

The Ohio State University, B.S., Computer Science and Engineering

December 2024